

Data Flow Diagram

Date	25 September 2026
Team ID	SWTID-2026-7436
Project Name	AI StudyBuddy API
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

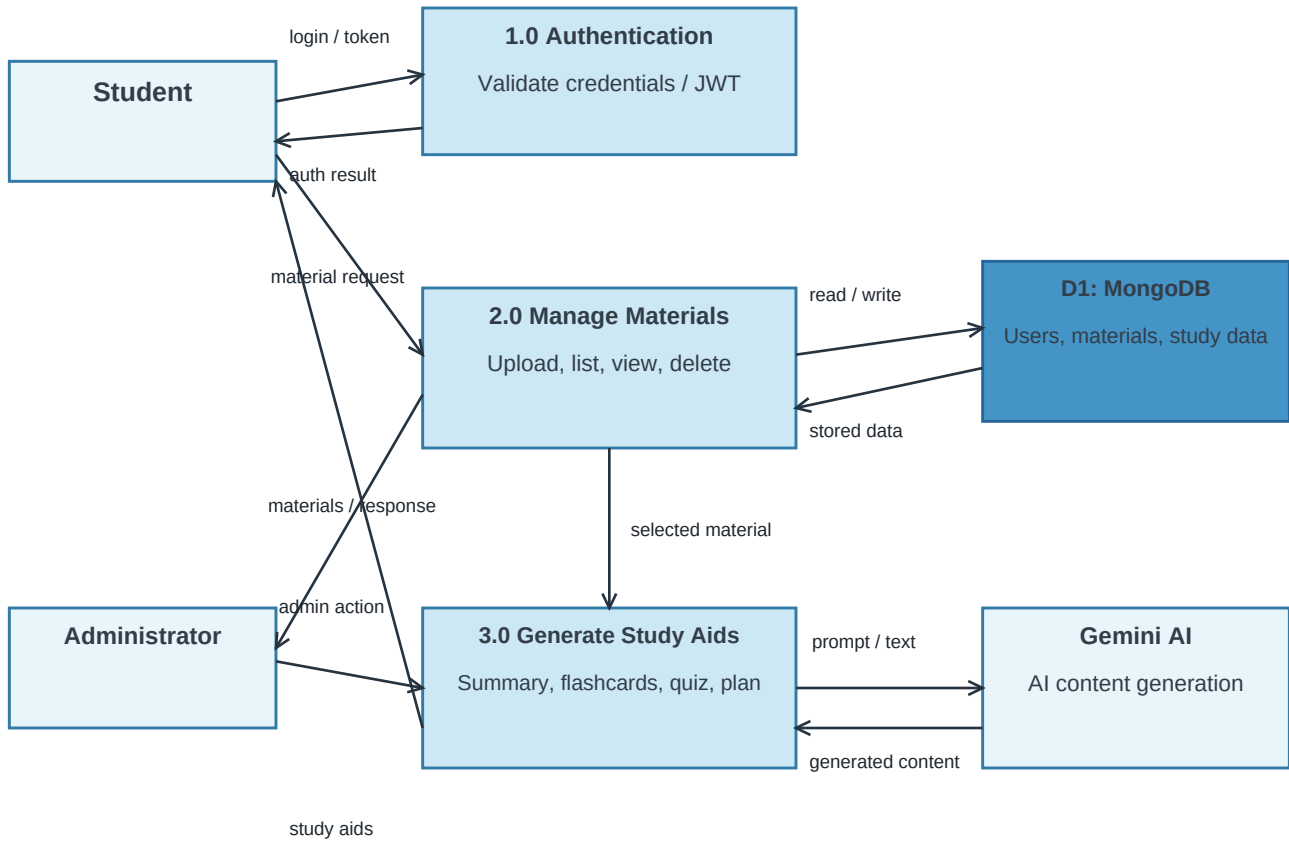
Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

AI StudyBuddy API - Data Flow Diagram

Student/Admin requests · API processes · database and Gemini AI



Data store: MongoDB (users, study materials, summaries, flashcards, quizzes, study plans).